SkyRescue: A UAV-Image-Based Decision Support System for Search and Rescue

Minh Hoang Tran[1*†], Ngoc Thao Anh Tran[1†], Thai Bao Le[1†], Nguyen Phuong Nghi Dang[1†], Minh Triet Le[1†]

[1*]Department, Organization, Street, City, 100190, State, Country.
[2]Department, Organization, Street, City, 10587, State, Country.
[3]Department, Organization, Street, City, 610101, State, Country.

*Corresponding author(s). E-mail(s): tmhoang1902@gmail.com;
†These authors contributed equally to this work.

Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

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1 Introduction

After earthquakes, floods, fires and major accidents, Search and Rescue (SAR) coordinators must decide which incidents need attention first and how to spend a limited number of teams. Unmanned aerial vehicles (UAVs) can photograph sites that are dangerous or hard to reach on foot. Those photographs help only if they can be reviewed quickly and consistently and if what they show reaches the people making response decisions. SkyRescue is designed to reduce manual image review and to support incident coordination. These are design goals. We did not measure response time or rescue outcomes.

Seeing the damage does not settle what to do about it. Coordinators read incident-specific indicators, compare reports of different urgency, and look for teams whose

skills and availability fit the need. Multi-criteria decision-making (MCDM) gives a structured way to compare alternatives against several criteria at once. Model outputs, meanwhile, can be wrong or can misrepresent real conditions, especially when evaluation relies on synthetic images or proxy indicators. SkyRescue therefore shows its results to the coordinator for review, and a prediction never authorizes dispatch.

This paper presents SkyRescue, a prototype that links UAV-image analysis to incident prioritization and rescue-team recommendation under coordinator control. The workflow validates images and metadata, classifies the scene, assesses severity by incident type, detects people, ranks incidents and recommends teams. The scene classifier assigns one of five classes: Collapsed Building, Fire, Flood, Traffic Accident or Normal. Which severity branch runs depends on the predicted class and on how confident the classifier is. A Normal report with high confidence skips severity scoring but stays available for the coordinator to check. Uncertain or unclassified outputs are held for review and are not treated as confirmed incidents. Human detection runs in parallel and passes person locations and posture information to the branches that use them.

For prioritization, SkyRescue combines the Analytic Hierarchy Process (AHP) and the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS). AHP derives criterion weights from pairwise comparisons [1]. TOPSIS ranks alternatives by their relative distance from the positive and negative ideal solutions [2]. After the coordinator verifies a non-Normal incident, team matching scores the eligible teams on specialization, distance and capacity, subject to operational constraints. Up to two support teams are added, and only when the primary team cannot meet the estimated demand. The system never dispatches on its own: the coordinator verifies the incident, chooses the assignments and confirms dispatch.

The paper makes three contributions. First, it specifies a workflow that links UAV-image intake and visual analysis to incident prioritization and coordinator-controlled response planning. Second, it organizes incident-specific assessment and human-detection outputs as structured evidence for later decisions and states the limits of the synthetic data and heuristic indicators behind them. Third, it defines a human-supervised recommendation process in which a primary team is kept apart from conditional support teams.

SkyRescue is a prototype. Its evaluation uses individual models, synthetic compositions and simulated team information, because no ground-truthed SAR deployment was available. The study reports component-level performance and the system design. Field validation and better rescue outcomes are outside its claims.

2 Related Work

2.1 UAV-Based Aerial Scene Classification for Emergency Monitoring

UAV imagery gives an aerial view of affected areas and can show visual patterns that go with emergencies. Kyrkou and Theocharides [3] proposed EmergencyNet, an efficient aerial-scene classifier for drone-based emergency monitoring. AIDER supplies

labeled aerial images of emergencies and has been used to develop scene-classification models [4]. Our classification component starts from these two works.

SkyRescue classifies scenes with EfficientNet-B0 [5] and does not use the EmergencyNet architecture. The classifier assigns each incoming image to one of five project-defined classes: Collapsed Building, Fire, Flood, Traffic Accident or Normal. Its output is the initial incident category, which sets the assessment route. The architecture, training configuration and preprocessing are described in the Methods section.

2.2 Human Detection in Disaster Scenarios

Detecting people in UAV imagery is hard. A person may cover only a few pixels or lie in an unusual posture, and terrain or nearby objects can hide part of the body. C2A, short for Combination to Application, is a synthetic benchmark for human detection in disaster scenes [6]. Its backgrounds come from AIDER, and its human figures come from the LSP and MPII-MPHB datasets. The figures are adjusted and composited onto the backgrounds, which gives controlled examples for training and evaluation. Synthetic composition cannot fully reproduce how people look in real rescue imagery.

Studies on real aerial imagery add evidence of a different kind. Gotovac et al. [7] evaluated visual person detection for SAR missions with UAS imagery and compared algorithmic detection with human interpretation. B"uttner et al. [8] released AIResQ, a high-resolution thermal dataset for airborne person detection, together with a benchmark assembled from drone-flight exercises involving SAR organizations. Their data reflect actual search conditions, such as small targets and unusual poses on varied terrain. The sensors, environments and evaluation setups in these studies differ from the synthetic RGB compositions in C2A.

C2A reuses AIDER backgrounds, so the two datasets are not independent by construction. Shared provenance does not by itself show that leakage occurred, but we have not yet audited background overlap across the datasets, and split-level checks are still needed. For that reason the current evaluations are not independent cross-dataset validation.

2.3 Incident-Specific Disaster Assessment

Datasets and methods for visual analysis tend to specialize by disaster type. For wildfire monitoring, FLAME2 offers UAV-acquired RGB and thermal imagery recorded during a prescribed forest-fire experiment [9]. Because the imagery comes from an experiment, it does not cover uncontrolled wildfire events. For post-earthquake building assessment, UAVs-TEBDE supports building-damage estimation [10]. It draws on several sources, so it should not be described as the product of a single UAV field campaign.

These resources do not show that every disaster type needs its own model, and image-level recognition does not fix an incident's priority. Priority also depends on the assessed consequences, on the response resources available and on the coordinator's judgment.

SkyRescue has assessment branches for Fire, Collapsed Building, Flood and Traffic Accident. The Fire and Collapsed Building branches use specialized visual-analysis methods built on MobileNetV3-Small [11]; software and model versions are listed in Methods. The Traffic branch adapts Fruin's [12] pedestrian level-of-service concept as a heuristic reference for estimated pedestrian density. Fruin's scale was developed for pedestrian-flow analysis and is not a standard for accident severity. The Flood branch uses the apparent height-to-width ratio of an upright person as a proxy indicator of severity.

Neither proxy is a physically calibrated measure. C2A has no ground-truth flood depth, and its composited figures cannot validate real immersion geometry. A pedestrian-density estimate from synthetic scenes likewise says little about the real severity of a traffic accident. We therefore report the Flood and Traffic outputs as heuristic indicators evaluated under the available conditions. Results driven by the detector are reported separately from results that use ground-truth annotation boxes, since annotation boxes carry no detector error.

2.4 Multi-Criteria Decision-Making for Incident Prioritization

When resources are limited, responders must decide which locations or incidents to address first. The decision can depend on urgency, geographic conditions and resource needs. He and Jung [13] proposed a voting-TOPSIS method for prioritizing disaster-damaged areas for relief distribution, combined the judgments of several experts, and tested its feasibility on a real flood case. Their study shows that multi-criteria ranking can be adapted to disaster prioritization.

SkyRescue uses AHP and TOPSIS, introduced above, to order incident reports so that the coordinator reviews higher-priority cases first. Methods specifies the criteria and weight values and the ranking calculations, including normalization and consistency statistics.

An MCDM method does not validate its own criteria or scales. In SkyRescue the incident-specific branches produce indicators with different meanings and different evidence behind them. Ranking them together assumes that their severity scores are comparable enough for prioritization. SAR domain experts have not yet validated the AHP judgments, and the comparability assumption remains untested. The ranking is decision support and should not be read as a validated measure of incident severity.

2.5 Integrated Decision Support for Disaster Response

Decision support for disaster response covers more than single prediction models. Rye and Aktas [14] studied PREDIS, a disaster-management decision-support model that combines estimation of disaster impact and needs with supplier selection. They evaluated it in a participatory simulation with experts and non-experts, so the evaluation looked at how people use recommendations as well as at model output. Cavdur et al. [15] developed a decision-support tool for allocating temporary disaster-response facilities under uncertainty, and Rolland et al. [16] addressed the assignment and scheduling of personnel and assets for disaster response and recovery.

These systems start from different inputs. PREDIS works from impact estimates and needs assessment and ends in humanitarian supplier selection, facility-allocation tools work on relief-distribution infrastructure, and personnel-and-asset scheduling research assigns tasks under operational constraints. SkyRescue starts from UAV imagery and connects scene classification and incident-specific visual indicators to report prioritization and team recommendation. The difference is one of scope and workflow, and we make no claim that earlier systems lack integration or human oversight.

SkyRescue is a prototype that integrates UAV-based visual analysis with coordinator-controlled prioritization and team recommendation. Its contribution is the design of this workflow and a plain account of the evidence behind it: synthetic human-scene compositions, heuristic Flood and Traffic indicators, simulated team information and no ground-truthed field evaluation. We do not claim operational superiority or uniqueness.

3 Data and Method

3.1 System Overview and Architecture

SkyRescue is a decision-support system, demonstrated locally, that links UAV-image intake to incident assessment, prioritization, rescue-team selection and Coordinator-controlled dispatch. It has two web portals. The Reporter portal (`/reporter/`) accepts incident images and lets reporters follow their submitted reports. The Dispatch portal (`/dispatch/`) is used by the SAR Coordinator or administrator to inspect reports, review AI evidence, correct operational incident information, verify incidents, select a rescue team, confirm dispatch and close incidents. One FastAPI application origin serves the static frontends, the API endpoints and the machine-learning service.

The server makes the final decision on every upload. The browser can give preliminary validation feedback, but the upload API decides acceptance. It takes JPEG and PNG files up to 35 MB and checks their magic bytes before reading any EXIF metadata. EXIF carries standardized image metadata such as capture time, device information and GPS fields [17]. In UAV mode the API checks the source UAV, comparing the registered UAV record with the serial number and camera manufacturer read from the image. It also checks geographic and temporal constraints and detects duplicate content. EXIF GPS is the preferred location source. Browser GPS is a fallback only for personal uploads without EXIF GPS, and demo-mode metadata is the last fallback and must be labelled as simulated.

The active-region geofence is a bounding box with a 0.5-degree margin, not an administrative polygon. Capture time may be at most 30 minutes old and at most 60 seconds in the future. `SHA-256` hashes identify duplicate images. Each failed upload is recorded with an error code, and the API limits each IP address to 30 upload requests per minute.

After a successful upload, the server creates a report ID and stores the image under `media/` with a random 128-bit filename. A background job then processes the report. A semaphore serializes inference, so the local demonstration runs one inference job at a time, and the processing route times out after 60 seconds. Model 1 classifies

the aerial scene and Model 2 detects people. The incident router selects the severity branch, validates its output contract and stores the accepted prediction and person detections. Model 2 returns image-pixel coordinates only, so the system does not treat a bounding box as a victim's GPS position.

The frontends stay in sync with the server through Server-Sent Events at `/api/events` and a poll of `/api/version` every two seconds. The server is the authority for validation and for AHP/TOPSIS results. A JavaScript priority calculation may be shown as a preview but never replaces the server result. AI outputs are advisory: they do not verify an incident or dispatch a team, and the Coordinator reviews the evidence and authorizes every operational action.

A report has four persisted states: Unverified, Matching Team, In Progress and Resolved. A new report enters Unverified. For a non-Normal incident, Coordinator verification moves the report to Matching Team and enables detailed team matching. An Unclassified report cannot be verified until the Coordinator selects a supported incident type. A Normal report stays visible for review and can be closed through a Normal-specific path without dispatching a team. Once dispatch is confirmed, the selected team becomes On Mission and the report becomes In Progress. Finalize & Resolve then closes the report as Resolved, marks the dispatch Completed, returns the team to Available and writes an audit record.

[FIGURE 1 ABOUT HERE: Overall SkyRescue System Architecture. Draw one system-level diagram showing the Reporter and Dispatch portals; FastAPI validation and background ML service; Model 1 and Model 2; the four incident-specific severity branches; server-side AHP/TOPSIS; Coordinator review and verification; one-team matching; dispatch lifecycle; and SQLite, media storage and audit logging. Do not depict support-team coalitions or advanced modules that are absent from the current demonstration. Place at the end of Section 3.1.]

3.2 Datasets and Data Preparation

Four datasets serve four tasks. Their image sources, labels and evaluation questions differ, so they are kept apart and not merged into one benchmark.

AIDER supports aerial-scene classification. It holds 6,433 aerial images in five classes: Collapsed Building (511), Fire (521), Flood (526), Normal (4,390) and Traffic Accident (485) [4]. The split is stratified 70/15/15 with seed 42, giving 4,503 training, 965 validation and 965 test images. Normal accounts for 4,390 of the 6,433 images and for 658 of the 965 test images, so accuracy and class-wise results must be read with that imbalance in mind. AIDER was built for aerial emergency-response scene understanding, a task also studied in the drone-based EmergencyNet work [3].

C2A, the Combination to Application dataset, supports human detection. It has 10,215 synthetic disaster-scene images with 360,467 person bounding boxes [6]. Human figures from LSP and MPII-MPHB are composited onto disaster-scene backgrounds taken from AIDER. The original posture labels (Upright, Bent, Kneeling, Lying and Sitting) are mapped to two project labels, Upright and Other, where Other merges the four non-upright postures. Images are split by background with seed 42 into 7,150 training, 1,530 validation and 1,535 test images. Splitting by background reduces contextual overlap between subsets. C2A still reuses AIDER backgrounds, so

the two datasets are not independent by construction, and C2A performance is not independent cross-dataset validation against AIDER.

FLAME2 supports fire severity. It provides UAV-acquired RGB and thermal imagery from a prescribed-fire experiment [9]. The Fire branch uses an RGB subset of 2,800 images and no thermal frames: 718 No Flame/No Smoke, 750 Flame/No Smoke and 1,332 Flame/Smoke. Temporal blocks with buffer blocks form training, validation and test subsets of 1,960, 420 and 420 images. A prescribed fire gives controlled imagery and does not cover the range of uncontrolled wildfires.

UAVs-TEBDE supports collapsed-building severity. It holds 1,636 post-earthquake UAV images labelled Intact (498), Damaged (664) and Collapsed (474) [10]. A stratified 70/15/15 split with seed 42 yields 1,145 training, 245 validation and 246 test images. The dataset is multi-source and tied to one post-earthquake imaging domain, so its results do not carry over to every building type, camera geometry or disaster environment.

[TABLE I ABOUT HERE: Data Sources and Data Splits. Include one row each for AIDER, C2A, the FLAME2 RGB subset and UAVs-TEBDE. Columns: task, labels, total images or boxes, train/validation/test counts, split method and seed, and the relevant imbalance or domain caveat. Do not infer class counts within individual splits unless the split manifest or label files support them. Place in Section 3.2.]

3.3 Scene Classification and Human Detection

3.3.1 UAV Scene Classification

Model 1 assigns each accepted UAV image to Collapsed Building, Fire, Flood, Normal or Traffic Accident. It is an EfficientNet-B0 initialized with ImageNet weights and a five-output head [5]. The input is 224×224 pixels and the inference checkpoint is `aider_classifier.pth`.

Training starts with the backbone frozen and then unfreezes the final feature blocks and the head. Augmentation consists of RandomResizedCrop (scale 0.6 to 1.0), horizontal and vertical flips, rotation of up to $\pm 20°$, ColorJitter, Gaussian blur, sharpness adjustment and RandomErasing, followed by ImageNet normalization. Optimization uses Adam [18] with learning rates of 1×10^{-3} for the head and 1×10^{-4} for the backbone, weighted cross-entropy, a batch size of 32 and at most 15 epochs. ReduceLROnPlateau tracks validation macro-F1. Training ran on a Google Colab T4 GPU.

The router applies the Normal gate before the incident-class thresholds. When Normal is predicted with $P(\text{Normal}) \geq 0.90$, severity processing is skipped and the report stays in the Unverified queue for Coordinator review. When Normal is predicted below 0.90, the image goes to the most probable incident class with a review flag. For an incident-class prediction, confidence of 0.70 or more keeps the class, confidence from 0.50 up to 0.70 keeps the class with a low-confidence review flag, and confidence below 0.50 produces Unclassified. An Unclassified report stays visible but cannot be verified until a supported incident type is assigned. The top incident prediction may remain as a review signal and is not a confirmed classification.

A threshold diagnostic for the 0.90 Normal gate recovered all 11 incident cases that AIDER had misclassified as Normal, and it flagged 46 of 630 Normal examples (7.3%) unnecessarily. On C2A it recovered 44 of 60 such cases and missed 16. These counts describe threshold behavior on the stated datasets and are not a calibration guarantee. The 0.70 and 0.50 incident-confidence thresholds are initial operating values, and no systematic threshold sweep has been run.

3.3.2 Human Detection and Posture Classification

Model 2 detects people and gives each one the project posture label Upright or Other. It is Ultralytics YOLOv8m, version 8.4.171, initialized from `yolov8m.pt` and run at 640×640 pixels [19]. The inference checkpoint is `v8m_up2_best.pt`, with a confidence threshold of 0.25 and an IoU threshold of 0.50. Each detection holds a pixel-space bounding box (x1, y1, x2, y2), a confidence score and a posture label. The coordinates place a detection within the image and are not geographic coordinates.

Training on C2A used 40 epochs, batch size 16, image size 640, automatic optimizer selection, initial and final learning rates of 0.01, momentum 0.937, weight decay 5×10^{-4}, three warm-up epochs, mosaic augmentation of 1.0, a horizontal-flip probability of 0.5, `close_mosaic` of 10, seed 0 and automatic mixed precision. Training took about 2.9 hours on Colab. The available metrics come from the C2A validation split and are reported as validation metrics, not test performance.

[TABLE II ABOUT HERE: Configuration of the Four Core AI Models. Include EfficientNet-B0, YOLOv8m, Fire MobileNetV3-Small and Collapsed MobileNetV3-Small. Report input size, dataset, output labels, checkpoint, optimizer and training configuration, and the evaluation split available for each. Traffic and Flood must stay described as heuristic or proxy branches and not as separately trained models. Place after Section 3.3.2.]

3.4 Incident-Specific Severity Assessment

SkyRescue sends each non-Normal incident report to one of four severity branches. Fire and Collapsed Building use trained image classifiers. Traffic and Flood use heuristic indicators derived from detector output. The four outputs differ in meaning and in the evidence behind them, so none of them is a ground-truth severity measurement that can stand in for another.

3.4.1 Fire Severity

The Fire branch is a MobileNetV3-Small with ImageNet initialization that classifies three states: No Flame/No Smoke, Flame/No Smoke and Flame/Smoke [11]. It is trained on the 2,800-image RGB subset of FLAME2 with weighted cross-entropy, batch size 32 and a two-phase AdamW procedure [20]. The second phase unfreezes the final three feature blocks. The checkpoint is `best_fire_mobilenetv3_phase2.pth`. The three states map to branch scores of 0.0, 0.5 and 1.0. This is an ordinal convention of the decision-support layer. The values are not probabilities of injury, loss of life or fire spread, and they are not a calibrated measure of fire severity.

3.4.2 Collapsed-Building Severity

The Collapsed Building branch is a second MobileNetV3-Small with ImageNet initialization that classifies Intact, Damaged and Collapsed [11]. Its convention maps the classes to 0.0, 0.5 and 1.0. Training on UAVs-TEBDE uses weighted cross-entropy, AdamW with a learning rate of 1×10^{-4}, batch size 32, up to 30 epochs and patience 8 [20]. The inference checkpoint is `collapsed_building_best_v2.pt`.

The current rule sets `needs_review` when $P(\text{Collapsed}) \geq 0.04$, and the earlier 0.75 confidence rule is disabled. The 0.04 threshold was tuned on the test set and has not been validated independently, so the reported review behavior and recall may be optimistic. The router also flags a zero severity output for a Fire or Collapsed report for Coordinator review, because it may signal a mismatch between the predicted incident class and its branch output.

3.4.3 Traffic Severity

The Traffic branch estimates pedestrian density from Model 2 detections. Detections with confidence of at least 0.25 are reduced to bounding-box centres and converted from pixels to metres with a ground-sampling-distance (GSD) assumption. GSD depends on flight altitude, focal length and sensor geometry [21]. The calibration assumes an altitude of 12 m, a 6.4×4.8 mm sensor and a 4.5 mm focal length, and trustworthy altitude metadata takes precedence when it exists.

A 4×4 m window moves over the estimated ground plane in 1 m steps to find the densest region. Density is the number of detected people per 16 m^2. The estimate is mapped to Fruin pedestrian level-of-service (LOS) grades A to F, which give branch scores from 0.0 to 1.0 in steps of 0.2 [12]. Fruin's scale was developed for pedestrian flow and not for accident severity. The Traffic output is therefore a pedestrian-density proxy that can help the Coordinator read crowd concentration. It is not a validated traffic-accident severity label.

3.4.4 Flood Severity

The Flood branch uses a detected Upright person as a visual scale reference. A detection is eligible when its width is at least 8 pixels, its height at least 30 pixels and its confidence at least 0.40. The reference baseline is 2.0667, estimated from 488 reference boxes across 304 C2A training images. For each eligible detection i, the branch computes the body-proportion proxy $s_i = \text{clip}\left(1 - \frac{h_i/w_i}{2.0667}, 0, 1\right)$ and takes the maximum, $s_{\text{flood}} = \max_i(s_i)$.

When no eligible Upright detection exists, the branch assigns the maximum score of 1.0 and sets `needs_review`. This keeps a missing reference person from reading as low flood severity. It can also over-prioritize reports and add to the Coordinator's review workload. The score is an image-based body-proportion proxy. It is not flood depth, a water-level measurement or calibrated immersion geometry.

3.5 Incident Prioritization Using AHP and TOPSIS

Once severity and report data are available, the server uses the Analytic Hierarchy Process (AHP) to derive criterion weights from pairwise judgments and the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) to rank the eligible incident reports in the current batch [1, 2]. The implementation has three criteria:

- Severity score (benefit). Higher severity raises priority. A Coordinator correction is used when one exists, and the AI-derived severity is used otherwise. The AI evidence stays available for traceability.
- Distance to the nearest qualifying team (benefit). Greater distance raises priority. This is a deliberate policy to give more attention to areas far from suitable resources, and it is not a travel-time minimization rule.
- Number of free qualifying teams (cost). Fewer free teams raise priority, because fewer available teams with the required specialization and coverage point to a resource shortage.

The AHP weights are 0.648 for severity, 0.230 for distance and 0.122 for the number of free qualifying teams, with a consistency ratio of 0.003. The development team set the pairwise judgments. They were not learned from rescue outcomes, and no SAR expert has validated them. The weights and the ratio describe the current configuration and are not empirically optimized policy parameters.

The server vector-normalizes the decision matrix, applies the AHP weights, finds the positive and negative ideal solutions from the criterion directions and computes each incident's relative distance to both. The closeness coefficient is the priority score, in the interval $[0, 1]$. It is relative to the current batch and is not a probability of harm or a calibrated measure of risk. The ranking covers active emergency incidents, verified or not. Normal and Unclassified reports stay visible in the Coordinator queue and are left out of the ranking. A `needs_review` flag does not add to or subtract from the TOPSIS score.

The nearest qualifying team is chosen among teams marked Available whose specialization and coverage fit the incident. The specialty mapping is Fire to fire, Flood to water_rescue, Collapsed Building to urban_sar and Traffic Accident to traffic_rescue. Coverage rests on explicit `TeamCoverage` records and a 10 km radius rule. Distance is the Haversine distance from the report GPS location to the team's headquarters. When no qualifying team exists, distance is NULL. For the TOPSIS calculation only, the server then substitutes the maximum distance in the current batch and sets `no_team_flag`.

This substitution has a side effect under the current directions. An incident with no qualifying team receives the maximum distance value and also has zero free teams, and both push it up the ranking, which can inflate its rank. The imputation and the criterion directions are policy choices, and they need a sensitivity analysis. Favoring distant incidents may support geographic equity, but it can conflict with minimizing response travel time. The conventional directions (distance as cost, free-team count as benefit) have not been compared with the current ones, so we do not claim the current directions are preferable.

The Python server computes the authoritative priority values, and the client-side formula is only a preview. At dispatch confirmation the server stores its priority snapshot with the dispatch evidence and includes it in the Digital Evidence Dossier, which can be exported as CSV or PDF.

3.6 Rescue-Team Matching

Detailed matching starts only after the Coordinator verifies a non-Normal incident. The current demonstration selects one primary team per report. It does not form support-team coalitions or assign several support teams. Eligibility is checked before the weighted ranking. The matching score combines specialization (0.50), distance (0.35) and capacity (0.15), and in the current specification capacity is a team's headcount and not a number of sub-units.

The rule engine has hard constraints and soft context bonuses, and hard constraints take precedence. HC-02 excludes a heavy truck from a narrow entrance. HC-03 requires a rescue boat for deep-flood or riverine-canal situations. HC-04 requires hydraulic heavy-debris equipment for a collapsed-building incident with severity of 88 or more. Among the seeded soft constraints, SC-01 favors an agile vehicle (+12 for Fire, +10 for Building Collapse and +4 for Flood), SC-03 adds +8 for a same-area team in an ultra-dense context, and SC-04 adds +10 for heavy-debris equipment in Building Collapse and +8 in Traffic Accident. When required context is missing, the evaluator records a soft warning and does not treat the missing value as a confirmed condition.

The operational notes define a demand convention of $\max(4, 2\,\text{people})$, where people is the detected or Coordinator-reported count. The plan does not make this estimate a separate hard eligibility gate, so it is described as a convention and not as an exclusion rule. A deadlock override path exists, and its reason and the responsible actor are recorded in Dispatch and the Audit Log.

[TABLE III ABOUT HERE: Decision-Support Configuration. Consolidate the AHP criteria, benefit/cost directions, weights and CR; the team-match weights; the headcount meaning and demand convention; HC-02/03/04 and SC-01/03/04; and the audited override path. Do not add a support-team column. Place after Section 3.6.]

3.7 End-to-End Workflow, Coordinator Control, API and Database

The integrated workflow begins with image upload and server-side validation. It continues through report creation, model inference, severity routing, AHP/TOPSIS ranking, Coordinator review and verification, one-team matching and dispatch confirmation. The server exposes endpoints for state retrieval and synchronization (`/api/state`, `/api/sync`), event and version updates (`/api/events`, `/api/version`), uploads (`/api/uploads`), prediction and rerun operations (`/api/ml/predict`, `/api/ml/rerun`), priority calculation (`/api/ahp/priority`) and service health (`/api/health`). The two portals use these routes, and the authoritative operational state stays on the server.

The Coordinator can review the original image, the classification and detection evidence, the severity output, the review flags, the priority information and the team

recommendation. Before verification, the Coordinator may correct the incident type, the severity and the victim count. An Unclassified report cannot pass the verification gate until a supported incident type is selected, and a Normal report has its own close path with no dispatch. For a verified non-Normal report, the matching module selects one eligible primary team. Confirm Dispatch sets the team to On Mission and the report to In Progress. Finalize & Resolve sets the report to Resolved, marks the dispatch Completed, returns the team to Available and writes an append-only audit record. The workflow has no separate rescue-result form and no field for the number of people recovered.

The persistent backend is SQLite in relational mode. The database is stored at `data/skyrescue.db` with write-ahead logging (WAL) and foreign-key enforcement enabled [22]. `DATA_BACKEND=relational` is the default and JSON is a fallback backend. JSON and camelCase representations are also used to project server state to the frontends, but the system is a relational SQLite application and not a document store. Uploaded images stay in the media directory and are not stored as database blobs.

The relational model has 15 domain tables: users, uavs, locations, `rescue_teams`, `team_coverage`, reports, predictions, `person_detections`, dispatches, rules, tags, `entity_tags`, `system_config`, `audit_log` and `upload_attempts`. Four infrastructure tables (sessions, `media_uploads`, `ml_results` and `app_meta`) hold authentication and session state, media processing, ML outputs and application metadata. A Report has one Prediction (`report_id` is unique), and a Prediction can have several PersonDetection records, one per bounding box. A Report has one Dispatch in the current demonstration (`report_id` is unique in Dispatch). The workflow therefore covers one incident and one primary team, and it has no multi-report evidence fusion and no coalition model.

The data model stores AI-originated values and operationally corrected values separately, and a Coordinator edit never overwrites an AI output. The `priority_score` is a TOPSIS closeness value in $[0, 1]$ and is distinct from the incident severity scale. The `no_team_flag` and the model-version fields are kept for traceability. A Dispatch record holds the selected team, the match score, the rule evaluation, the override information and the AHP priority snapshot. Audit Log entries are append-only and tie each action to the authenticated server session.

4 Results and Discussion

4.1 Experimental Setup and Evaluation Protocol

Results are reported separately for each model, dataset and kind of evidence. Three learned branches have held-out test metrics: EfficientNet-B0 on AIDER, the Fire MobileNetV3-Small on FLAME2 and the Collapsed Building MobileNetV3-Small on UAVs-TEBDE. YOLOv8m has validation metrics on C2A and no test metrics. Traffic is evaluated with a consistency analysis, and Flood is summarized through review and fallback behavior. AHP/TOPSIS and team matching need reproducible decision-support scenarios. System integration is assessed with automated tests and a local end-to-end smoke test. These kinds of evidence are not interchangeable and are not combined into an overall system accuracy.

Scene classification is reported with accuracy, macro precision, macro recall and macro-F1. Fire and Collapsed Building are reported with test accuracy, weighted F1 and class-wise values where available. Object detection uses precision, recall and mean average precision, with mAP50 and mAP50–95 following the COCO evaluation convention [23] as implemented in Ultralytics [19]. The four core AI models were trained on Colab T4, and the YOLOv8m run took about 2.9 hours. System timing was measured separately on one local Docker/CPU/WSL2 environment, and those figures apply only to that runtime.

4.2 AI Model and Severity Results

[TABLE IV ABOUT HERE: Consolidated AI and Severity Results. Include six rows: scene classification, human detection, Fire, Collapsed Building, Traffic and Flood. Columns: dataset, split or evidence type, metric, reported value, and interpretation or caveat. Mark YOLOv8m as validation only, Traffic as consistency analysis and Flood as review and behavior statistics. Place at the beginning of Section 4.2.]

4.2.1 Scene Classification

EfficientNet-B0 reached 95.2% accuracy on the 965-image AIDER test split, with 919 correct predictions. Macro precision was 0.912, macro recall 0.945 and macro-F1 0.927. The test split has 658 Normal images (about 68.2%), so a model that always answered Normal would score 68.2%. The reported accuracy is 27.0 percentage points above that baseline. The imbalance still makes the class-wise results and the confusion matrix necessary for a full reading. The result supports scene classification on the held-out AIDER split. It does not show performance on operational UAV images with other cameras, altitudes, weather or disaster conditions [4].

4.2.2 Human Detection

On the C2A validation split, YOLOv8m scored precision 0.415, recall 0.692, mAP50 of about 0.454 and mAP50–95 of 0.302. The mAP measures follow the COCO evaluation convention [23]. The model found part of the annotated people and also produced many false positives. The gap between recall and precision affects the downstream branches: false detections can inflate person counts or supply unsuitable upright references, and missed small people shrink the evidence available to Traffic and Flood assessment.

These are validation results and not test results. C2A consists of synthetic composites built from AIDER backgrounds and figures from other datasets [6], and it cannot fully reproduce the small targets, occlusion, unusual postures, terrain, sensors and search conditions of real UAS missions. Studies of person detection on real UAS imagery and the thermal AIResQ dataset cover those conditions [7, 8]. The detector needs evaluation on the C2A test split and on real UAV imagery before any claim about field detection performance.

4.2.3 Fire Severity

The Fire classifier reached 93.57% accuracy and a weighted F1 of 0.9352 on the 420-image FLAME2 test split. Validation accuracy and weighted F1 were higher, and

the test figures are the held-out result. The score applies to the RGB subset of one prescribed-fire experiment and does not extend to uncontrolled wildfires or other thermal and RGB acquisition settings [9]. The branch values of 0.0, 0.5 and 1.0 are ordinal scores mapped from three visual states and are not injury or life-safety probabilities.

4.2.4 Collapsed-Building Severity

On the 246-image UAVs-TEBDE test split, the Collapsed Building classifier reached a weighted F1 of 0.9512. For the Collapsed class, precision was 1.00 and recall 0.90. The review threshold $P(\text{Collapsed}) \geq 0.04$ was tuned on this same test set, so the threshold-dependent review result is exploratory, may be optimistic and is not an independently validated operating point. UAVs-TEBDE is multi-source and centred on post-earthquake building imagery, so its metrics do not carry over to other disaster regions or building domains [10].

4.2.5 Traffic Severity

The C2A consistency analysis gives 71% same-grade agreement, 98% agreement within one grade and a Spearman correlation of 0.73 between people count and the generated severity value. These are internal consistency indicators and not field accuracy. The Traffic score comes from estimated pedestrian density, so its correlation with people count does not independently validate real accident severity. Fruin's LOS scale offers a pedestrian-flow reference and no ground-truth labels for the consequences of traffic incidents [12]. Camera geometry and altitude assumptions also affect the pixel-to-ground-distance estimate [21].

4.2.6 Flood Severity

About 83.5% of the evaluated C2A images were flagged for review by the Flood branch. This is a review rate. It is not a classification accuracy and not a measure of flood-depth estimation. When no eligible Upright detection exists, the fail-safe rule assigns the maximum score and requests review. That is conservative about missing visual references, but it can push reports up the priority queue and add to Coordinator workload. The share of fallback cases and their effect on queue order have not been quantified yet.

[FIGURE 2 ABOUT HERE: AI Model Evaluation Panel. Composite panels: (a) EfficientNet-B0 training and validation curves; (b) AIDER test confusion matrix; (c) representative YOLOv8m detections showing successful cases and also false positives and missed people; and (d) YOLOv8m precision-recall or validation plot. Label the YOLO panel as validation unless a test evaluation has been run and logged. Place after the model-result subsections.]

[FIGURE 3 ABOUT HERE: Incident-Specific Severity Examples. Four panels showing (a) the three Fire states and their branch scores; (b) Intact, Damaged and Collapsed outputs; (c) a Traffic density map with bounding-box centres and the $4 \times 4\,\text{m}$ *sliding window; and (d) a Flood Upright-reference case and a no-reference fallback case. State whether each panel uses live detector boxes or annotation boxes, and for Traffic name the comparator*

used in the 71% / 98% analysis. Use system outputs, not fabricated illustrations. Place after Section 4.2.6.]

4.3 AHP/TOPSIS and Team-Matching Results

The configuration gives severity a weight of 0.648, distance 0.230 and the number of free qualifying teams 0.122, with a consistency ratio of 0.003. Severity and distance are benefit criteria and free-team count is a cost criterion. Under this policy, an incident with higher severity, a greater distance to a qualifying team or fewer free qualifying teams tends to rank higher. He and Jung [13] used TOPSIS to prioritize disaster-damaged areas for relief distribution. Their work is a precedent for the method and does not validate SkyRescue's criterion directions or weights.

The distance direction is an underserved-area policy. It can favor remote locations with little access to suitable resources, and it may conflict with minimizing travel time or reaching the most incidents quickly. Treating free-team count as a cost criterion prioritizes scarcity, which is a different aim from maximizing immediate response capacity. A consistency ratio of 0.003 shows that the pairwise judgments agree with one another and does not show that they are operationally correct. Expert validation and a sensitivity analysis are still needed [1, 2].

The demonstration inventory holds 20 incidents across seven provinces and 18 rescue teams. These are demonstration data and not a verified roster, so team locations, availability, coverage, equipment and incident timestamps say nothing about real field capacity. No ranking scenario or sensitivity run has been exported yet, and this section therefore reports no ranks and no sensitivity statistics. The planned scenario takes five to eight active incidents from one saved demonstration state. It reports each incident's type, severity source, nearest qualifying distance, free-team count, `no_team_flag`, TOPSIS closeness coefficient and rank, so that a reader can reconstruct the ranking from the saved inputs. It also follows one verified non-Normal incident through the matching engine: the candidate teams, the hard-constraint results, the applicable soft bonuses, the specialization, distance and headcount components, and the single primary team selected.

The planned sensitivity analysis compares the current directions with the conventional ones, the current weights with equal weights, and the three-criterion ranking with a severity-only ranking. It also perturbs each weight by ±20% and renormalizes. Rank correlation, top-k overlap and maximum rank shift will be reported once these runs are executed and logged.

[FIGURE 4 ABOUT HERE: AHP/TOPSIS and Team-Matching Scenario. Use one reproducible saved batch of five to eight demo incidents. Panels: (a) the TOPSIS ranking with input criteria and closeness scores; (b) sensitivity comparisons after the specified analyses have been run; and (c) candidate filtering and the rule evaluation leading to exactly one selected primary team. If no team is eligible, show the `no_team_flag` and the distance-imputation effect. Label all seeded inventory and GPS information as demonstration data. Place at the end of Section 4.3.]

4.4 End-to-End System Testing and Demonstration

The local demonstration has automated test evidence at three levels. The Python pytest suite reports 45 passed tests covering API behavior, Layer 1 validation, seed validation, the upload flow, ML output contracts, router thresholds and AHP logic. The Node test suite reports four passed tests covering AHP vectors, EXIF handling and formula checks. A Playwright run of `e2e_smoke.mjs` is recorded as PASS. It is a smoke test of the demonstration workflow. It is not a long-duration reliability test or a field acceptance test, and the run carries a warning that demo data on the target server may be reset.

Performance was measured on one local Docker/CPU/WSL2 environment. The Docker image is 2.23 GB, and model loading took about 4.0 seconds. RAM use was about 592 MiB at idle and 715 MiB after the end-to-end run. Full inference for Model 1, Model 2 and the selected severity branch took about 368 to 431 ms, and upload-to-AI-result time was about 0.8 seconds. Upload-to-Dispatch synchronization took about 104 to 394 ms through SSE and at most two seconds on the polling path. These single-machine figures depend on the local runtime, the hardware, model warm-up and the single-inference execution policy, and they are not deployment benchmarks.

[TABLE V ABOUT HERE: End-to-End Functional Test Summary. Include separate rows for upload validation, AI/severity output contract, AHP ranking, matching and Coordinator actions, Confirm Dispatch and lifecycle, relational persistence and audit trace, automated test suites, and single-machine performance. Distinguish the 45 passing pytest tests, the four passing Node tests and the Playwright E2E smoke-test PASS. Give the runtime context with every performance figure. Place in Section 4.4.]

[FIGURE 5 ABOUT HERE: End-to-End Workflow and Persistence Evidence. Build a composite screenshot set that uses the same incident and report ID throughout: (a) Reporter upload; (b) Dispatch queue and AI evidence; (c) Coordinator review and Verify; (d) AHP priority snapshot and one-team match; (e) Confirm Dispatch with the team On Mission and the report In Progress; and (f) the corresponding SQLite rows, Audit Log entry and Digital Evidence Dossier snapshot. Capture these from one fresh run and do not combine unrelated incident IDs into a synthetic workflow. Place at the end of Section 4.4.]

4.5 Discussion and Limitations

Several individual models score well on their held-out datasets, but each score belongs to its task. AIDER's scene-classification accuracy has to be read against its Normal-heavy class distribution. C2A gives a controlled synthetic setting for human detection, but its composited people and AIDER-derived backgrounds do not match real UAV search imagery [6]. The YOLOv8m metrics are validation values and are modest, especially precision and mAP, and the detector still needs an independent test and real-UAV validation. Real UAS search-and-rescue studies and thermal datasets such as AIResQ involve different target scales, sensors, terrain and postures [7, 8].

The Fire and Collapsed Building classifiers were tested on their own datasets and not on a common deployment benchmark. FLAME2's prescribed-fire imagery does not cover uncontrolled wildfires [9], and UAVs-TEBDE is multi-source and specific to post-earthquake building assessment [10]. The Collapsed review threshold was tuned on the test split, so its behavior may be optimistic. Traffic and Flood outputs are proxies

with no field ground truth. Traffic inherits assumptions from camera geometry and from Fruin's pedestrian-flow categories, and an inaccurate altitude or GSD changes the estimated density [12, 21]. Flood depends on whether Upright detections exist and on their geometry, and its maximum-severity fallback can inflate priorities and review demand.

The decision-support layer embeds policy choices. The development team assigned the AHP weights and no SAR expert has validated them. A low consistency ratio does not show that the weights match operational priorities. Giving greater priority to farther incidents and to incidents with fewer qualifying teams supports an underserved-area objective, but it may conflict with minimizing response time. TOPSIS scores are relative to the current batch, and the no-team distance imputation can raise a case's rank. The ranking is a configurable decision aid. It is not a validated measure of incident risk and not an optimal allocation policy.

The quality of the demonstration inventory limits the team recommendations. The seeded records do not establish the real availability, location, coverage, specialization or equipment of response teams. The timing measurements come from one Docker/CPU/WSL2 machine and say nothing about multi-user load or deployment-scale inference. The active-region geofence is a bounding-box approximation.

Security and data governance also limit operational readiness. In the current demonstration the Reporter role can read reports beyond its own submissions, and broad state keys remain writable. Real personal or operational data should stay out of a public deployment until role-based filtering and state-write authorization are tightened. Reverse geocoding through Nominatim sends report coordinates to an external service, which privacy notices and deployment policy must address. These are limits of the current demo configuration, and the system has not passed a security audit.

The results show that an integrated local prototype can link UAV-image processing with server-side prioritization and human-controlled dispatch. They do not establish field accuracy, operational superiority, shorter response time, better survival or better rescue outcomes. Those claims need ground-truthed field data and an evaluation with SAR personnel.

5 Conclusion

5.1 Summary

SkyRescue is a UAV-image-based decision-support system. It connects image intake and server-side validation with aerial-scene classification, person detection, incident-specific severity assessment, AHP/TOPSIS prioritization, rule-based matching to one primary rescue team, Coordinator verification, dispatch confirmation, and relational persistence with audit traceability. In the two-portal implementation, AI outputs supply evidence and recommendations, and verification and dispatch remain human actions.

5.2 Main Findings

EfficientNet-B0 reached 95.2% accuracy and 0.927 macro-F1 on the AIDER test split [4, 5]. The Fire branch reached 93.57% test accuracy and 0.9352 weighted F1 on the FLAME2 RGB test subset [9, 11]. The Collapsed Building branch reached 0.9512 weighted F1 on the UAVs-TEBDE test split, although its review threshold was tuned on that set [10, 11]. YOLOv8m has C2A validation metrics of precision 0.415, recall 0.692, mAP50 of about 0.454 and mAP50–95 of 0.302, and it has no independent test metrics [6, 19]. Traffic's 71% same-grade agreement, 98% within-one-grade agreement and 0.73 Spearman correlation are consistency indicators, and Flood's 83.5% is a review rate. Neither measures field severity accuracy.

The AHP weights are 0.648 for severity, 0.230 for distance and 0.122 for free qualifying teams, with a consistency ratio of 0.003. The criterion directions encode a preference for more severe, farther and more resource-constrained incidents. Local integration evidence consists of 45 passing pytest tests, four passing Node tests and a Playwright end-to-end smoke test recorded as PASS. The timing figures from the single Docker/CPU/WSL2 environment form an initial demonstration profile and not a deployment benchmark. All results stay separated by model, dataset, evidence type and runtime context and are not combined into an overall accuracy.

5.3 Limitations and Future Work

The first step is to evaluate YOLOv8m on the held-out C2A test split and on real UAV imagery, including small or occluded people and unusual postures [6–8]. The scene-classification confidence thresholds need systematic sweeps and a calibration analysis. The Collapsed review threshold should be chosen on validation data and assessed once on a separate test set.

Flood and Traffic need field-relevant ground truth. That means collecting flood-depth or immersion references and traffic-incident severity labels, and calibrating GSD against actual altitude and camera geometry. The Fruin-derived Traffic indicator should remain a pedestrian-density proxy until it is validated against incident outcomes [12, 21]. The frequency of Flood fallback, and its effect on queue rankings and Coordinator workload, also needs measurement.

SAR experts should review the AHP judgments and the criterion directions, followed by a sensitivity analysis of alternative directions and weights [1, 2]. Team locations, coverage, availability, equipment and headcount need checking against operationally verified inventories. On the system side, role-based access control needs strengthening, the approximate geofence needs replacing with a precise boundary, and external geocoding and location-data handling need to be written into privacy governance.

The system should then be evaluated with SAR Coordinators in structured usability and decision-time studies that examine whether the evidence and recommendations are understood and used appropriately. Rye and Aktas [14] offer a precedent in the participatory evaluation of decision-support recommendations. Integration with operational personnel and asset allocation is a further direction, following earlier

disaster-response scheduling research [16]. Benefits in response time or rescue outcomes can be claimed only after a controlled evaluation against operational baselines with outcome data.

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